

EXAM I

NUMERICAL ANALYSIS

(MATH 461) FALL, 2023

DUE: 11/09/2023

Problem 1. Show that if $k \geq 1$ is an integer, then the sequence $a_n = 10^{-n^k}$ converges superlinearly to zero but not of order α for any $\alpha > 1$.

Problem 2. Suppose $A \in \mathbb{R}^{n \times n}$ is invertible and $b \in \mathbb{R}^n$ and $k \geq 2$ is an integer. Give an efficient algorithm for computing the solution x to $A^k x = b$ using $\frac{2}{3}n^3 + \mathcal{O}(kn^2)$ FLOPs. You may assume the Gaussian elimination is available for use as `ge_lu(A)` (taking A as an input and returning L and U stored in A), that no pivoting (i.e., no row interchanging) is required, and that the function `ge_lu(A)` takes $\frac{2}{3}n^3 + \mathcal{O}(n^2)$ FLOPs.

Problem 3. How many distinct positive numbers can be represented in a floating point system using base $b = 10$, precision $t = 2$ and exponent range $L = -9, U = 10$?

Problem 4. Write, in pseudocode, an algorithm to compute the determinant of an invertible matrix $A \in \mathbb{R}^{n \times n}$ using LU factorization.

HINT: If P is a permutation matrix that swaps rows i and j , we have $\det(P) = -1$. Also, recall that $\det(MN) = \det(M)\det(N)$.

Problem 5. Consider *Steffensen's method*, an iterative root finding method given by $x_{k+1} = x_k - \frac{f(x_k)}{g(x_k)}$ where

$$g(x) = \frac{f(x + f(x)) - f(x)}{f(x)}.$$

Show that, in general, the method converges quadratically to a root of $f(x)$. This makes Steffensen's method a desirable alternative to Newton's method (which also has quadratic convergence) in situations where information about the derivative of f is unavailable.

Problem 6. Given a sequence of real numbers x_n and some $x^* \in \mathbb{R}$, suppose there is a constant $C > 0$ such that

$$|x_{n+1} - x^*| < C|x_n - x^*|^2$$

for all $n \geq 0$. Give a condition on $|x_0 - x^*|$ and C that is nontrivial (for example, something other than $x_n = x^*$ for all $n \geq 0$ or the like) and ensures $\lim_{n \rightarrow \infty} x_n = x^*$. Then, explain how your answer would need to be modified if we instead had the bound

$$|x_{n+1} - x^*| < C|x_n - x^*|.$$

Problem 7. Let $\tilde{x}_0, \tilde{x}_1, \dots, \tilde{x}_k$ be a permutation of $x_0, x_1, \dots, x_k$ (i.e. the same points, but in a potentially different order) and let f be a function. Show that the final divided difference for f at the points $x_0, \dots, x_k$ is invariant to permutations; i.e. show that

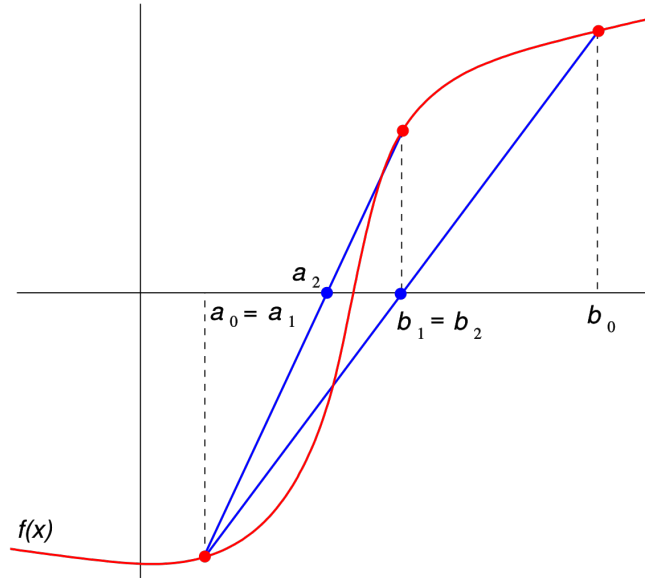
$$f[\tilde{x}_0, \tilde{x}_1, \dots, \tilde{x}_k] = f[x_0, x_1, \dots, x_k].$$

HINT: Consider the k^{th} derivative of the unique polynomial of degree at most k interpolating f at these $k+1$ points.

Problem 8. In the bisection method, the k^{th} approximation to the root, x_k , is given in terms of the k^{th} bracketing interval $[a_k, b_k]$ by the midpoint, i.e.;

$$x_k = \frac{a_k + b_k}{2}.$$

The *regula falsi* (Latin for *false position*) method is a modification to the bisection method where the estimated root, instead of being the midpoint of the bracketing interval, is replaced by the x-intercept of the line segment joining the graph of the endpoints of the function on the current bracketing interval. Visually, this looks like the following:



- (i) Derive the formula for the k^{th} approximation to the root, x_k , in terms of the k^{th} bracketing interval $[a_k, b_k]$.
- (ii) Compare your formula in part (i) of this problem to the secant method:

$$x_n = \frac{x_{n-2}f(x_{n-1}) - x_{n-1}f(x_{n-2})}{f(x_{n-1}) - f(x_{n-2})}.$$

Explain how the *regula falsi* method is different from the secant method despite this apparent similarity. In particular, when applied to a continuous function f and provided we begin with a bracketing interval $[a_0, b_0]$ where f has opposite signs at the endpoints, explain clearly why the *regula falsi* method will always converge, while the secant method may not.